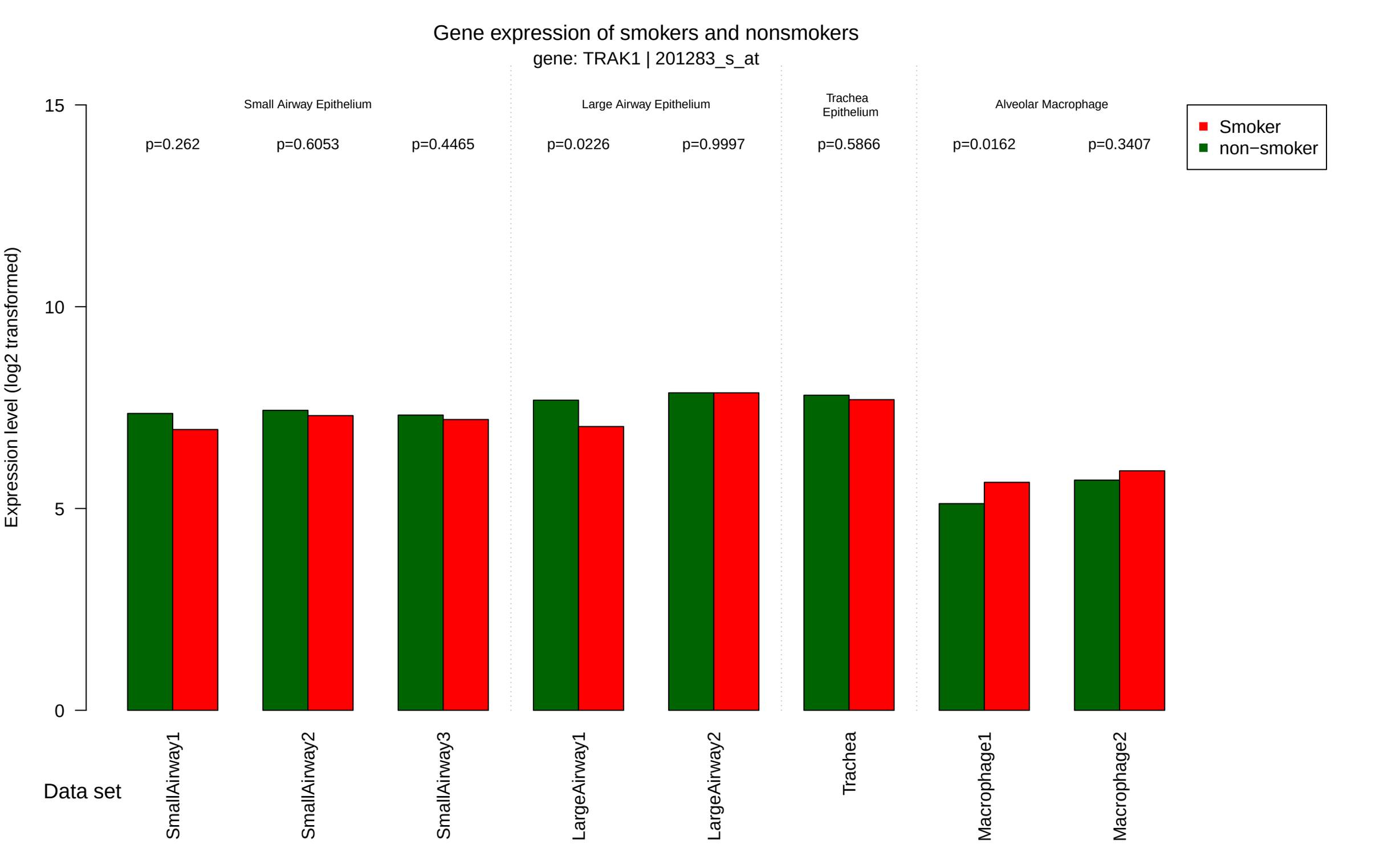
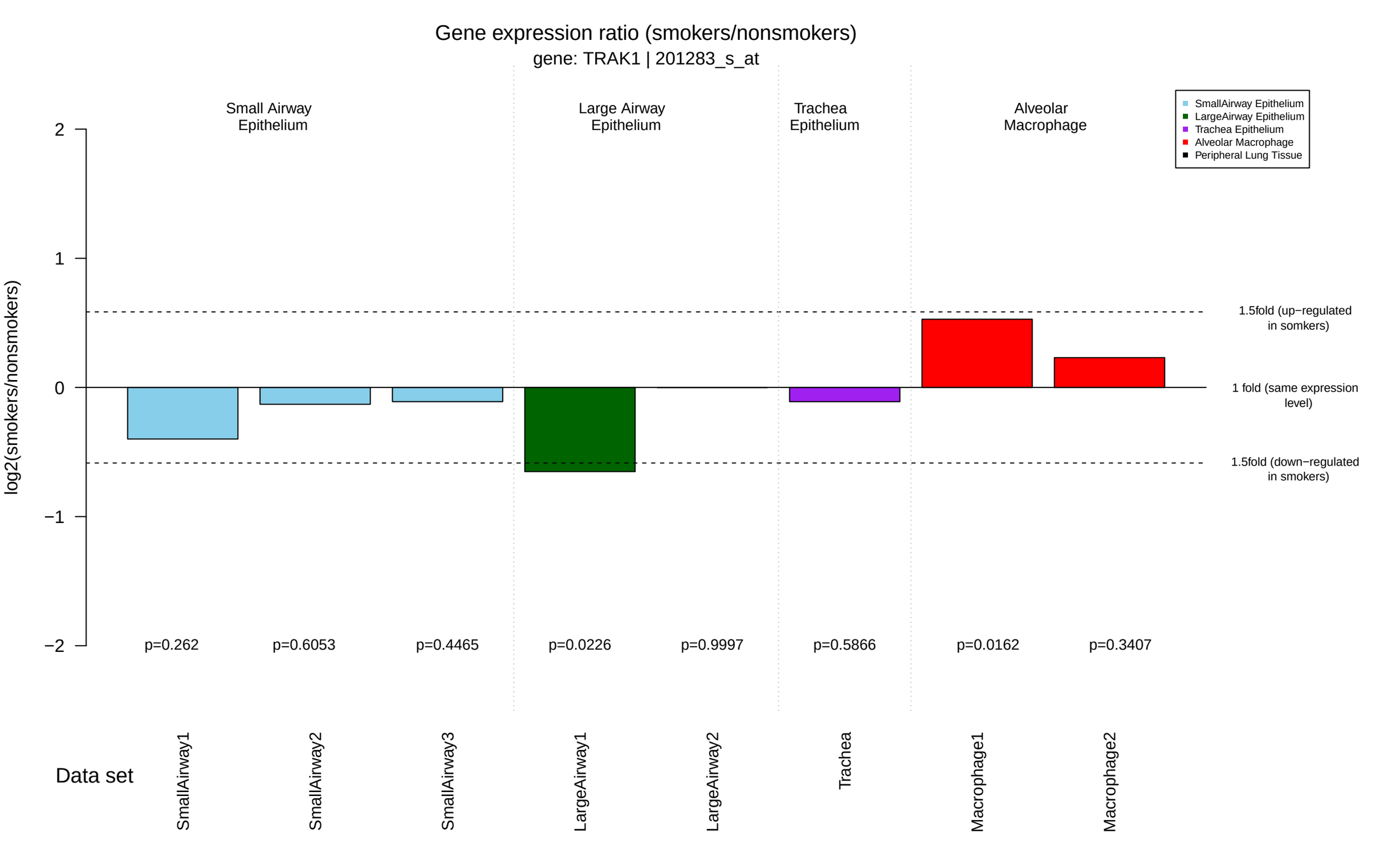
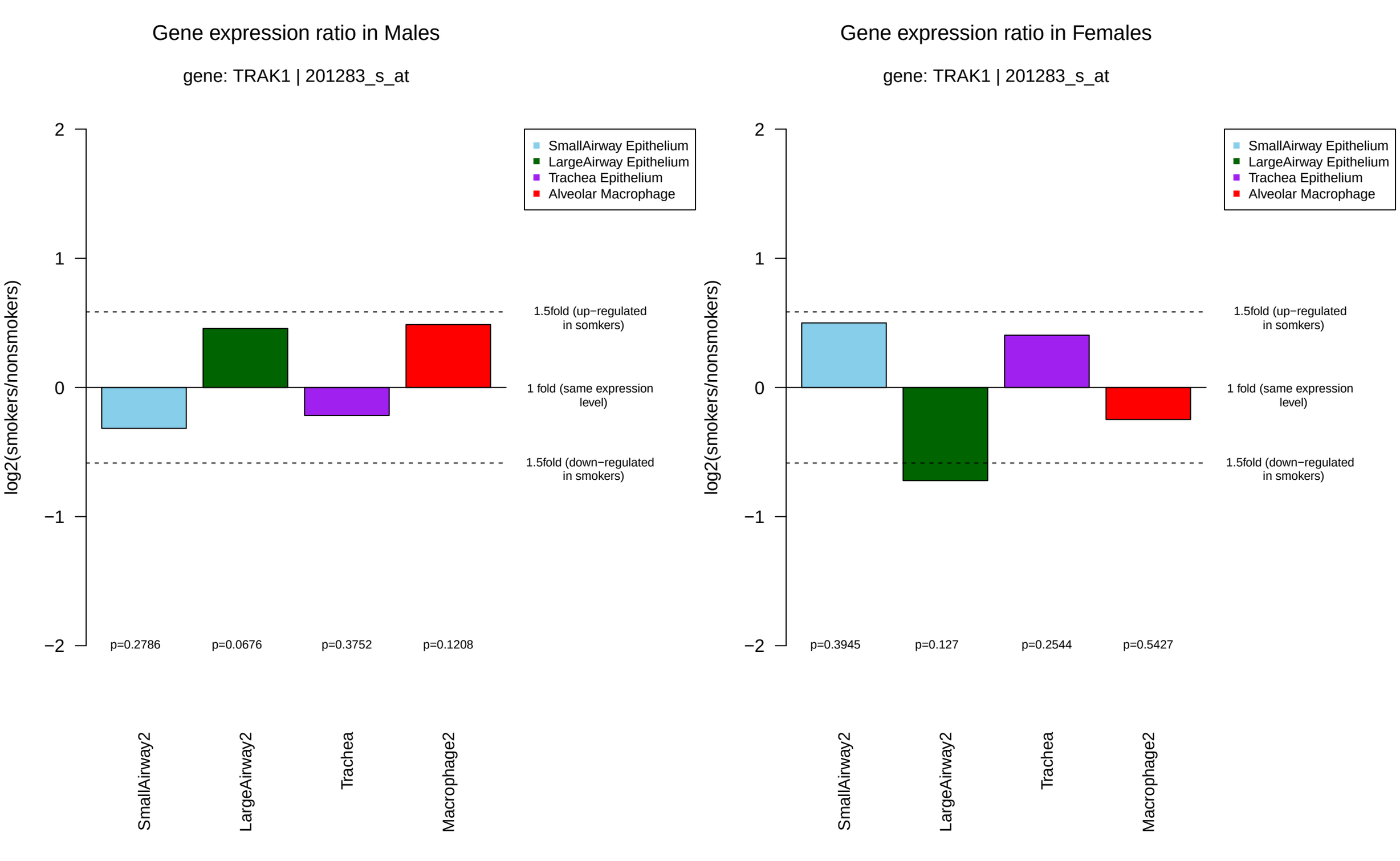
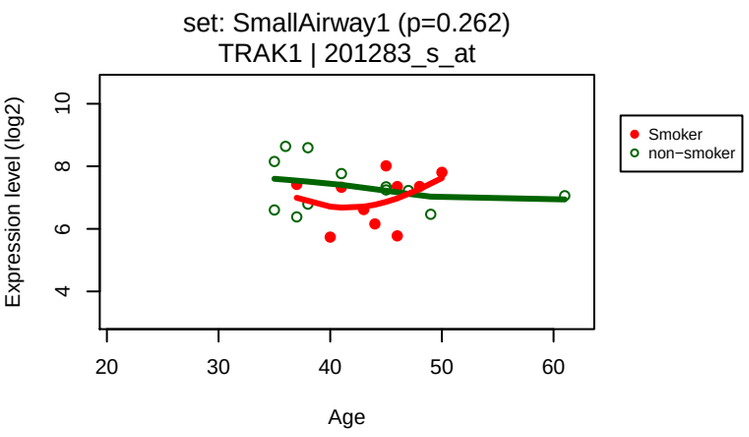
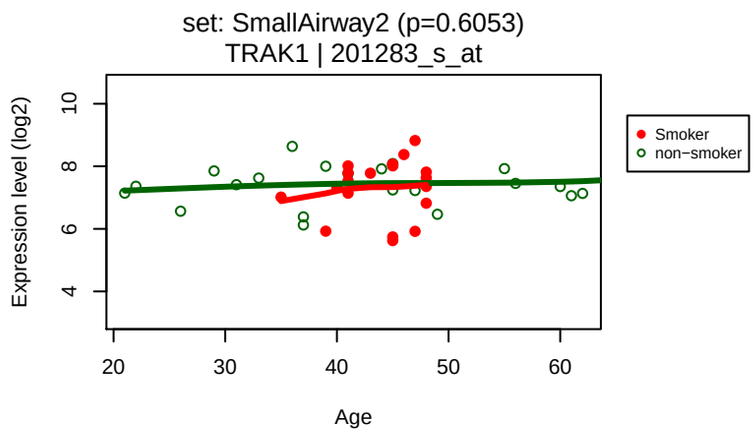
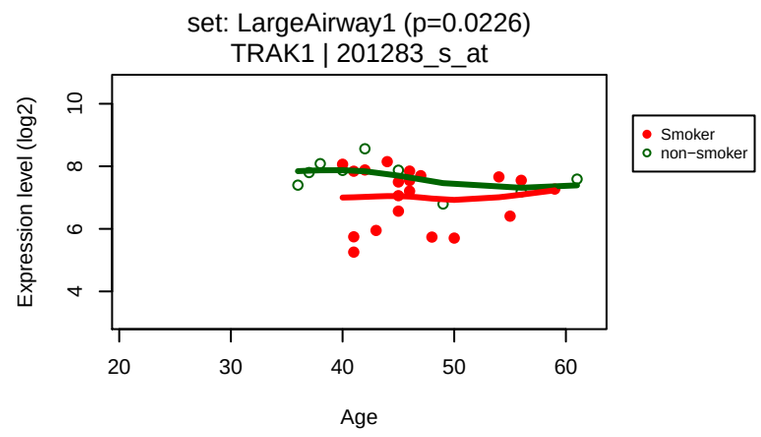

Gene expression of smokers and nonsmokers
gene: TRAK1 | 201283_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.262
p=0.6053
p=0.4465
p=0.0226
p=0.9997
p=0.5866
p=0.0162
p=0.3407
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

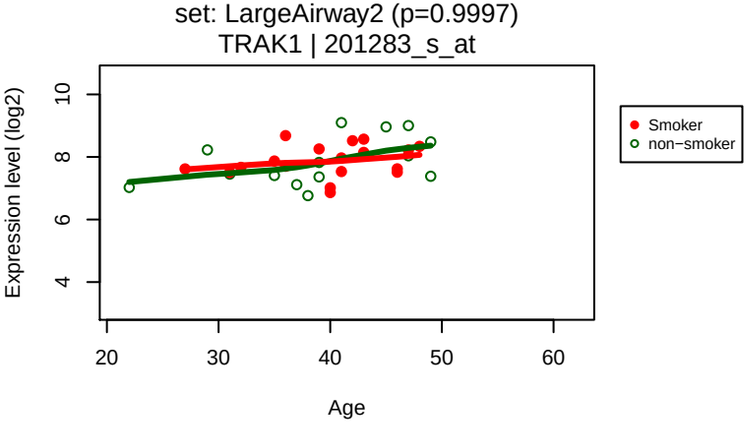

Gene expression ratio (smokers/nonsmokers)
gene: TRAK1 | 201283_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.262
p=0.6053
p=0.4465
p=0.0226
p=0.9997
p=0.5866
p=0.0162
p=0.3407
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

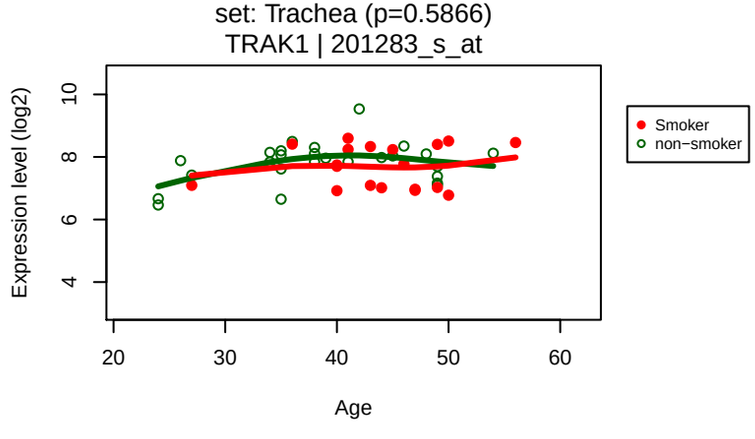

Gene expression ratio in Males
gene: TRAK1 | 201283_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2786
p=0.0676
p=0.3752
p=0.1208
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAK1 | 201283_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3945
p=0.127
p=0.2544
p=0.5427
SmallAirway2
LargeAirway2
Trachea
Macrophage2

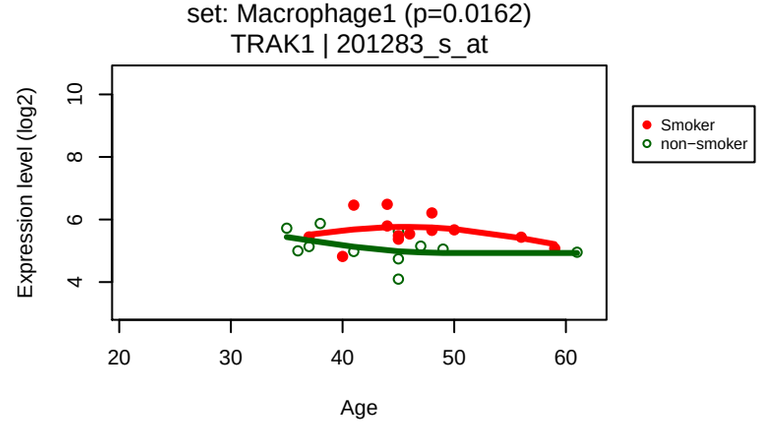

set: SmallAirway1 (p=0.262)
TRAK1 | 201283_s_at
Expression level (log2)
Age
Smoker
non−smoker

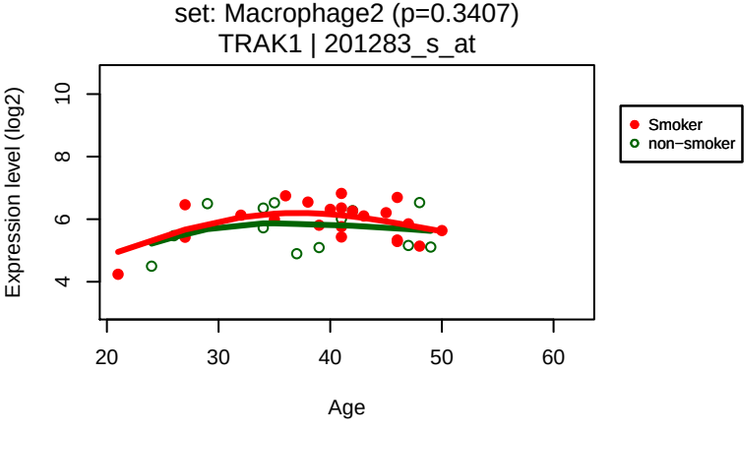

set: SmallAirway2 (p=0.6053)
TRAK1 | 201283_s_at
Expression level (log2)
Age
Smoker
non−smoker

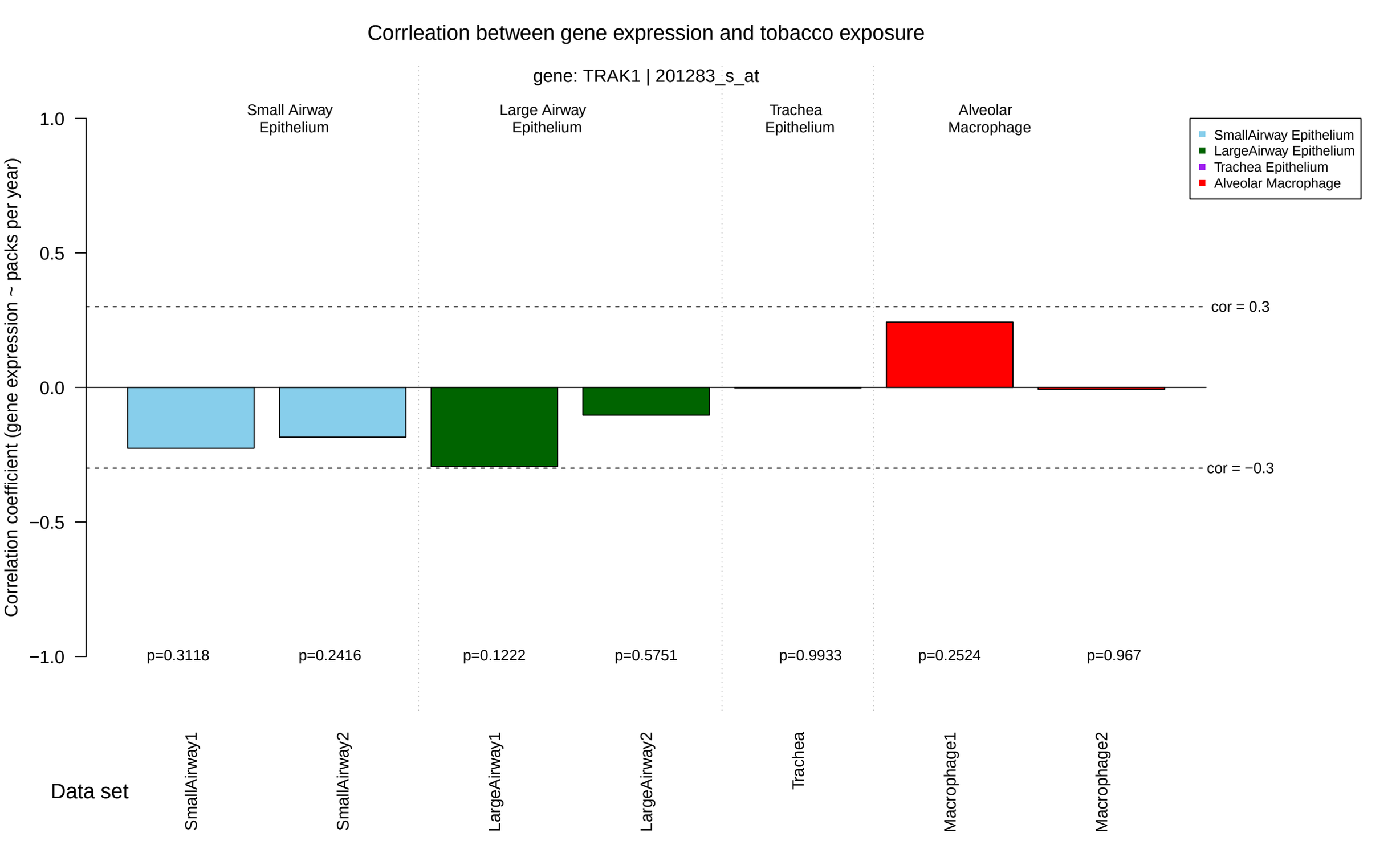

set: LargeAirway1 (p=0.0226)
TRAK1 | 201283_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9997)
TRAK1 | 201283_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5866)
TRAK1 | 201283_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0162)
TRAK1 | 201283_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3407)
TRAK1 | 201283_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAK1 | 201283_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3118
p=0.2416
p=0.1222
p=0.5751
p=0.9933
p=0.2524
p=0.967
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2